MsRE: Toward Efficient Remote Sensing Segmentation via Vision Foundation Models

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***Abstract*—Vision foundation models (VFMs) pretrained on large-scale datasets have significantly improved performance in remote sensing semantic segmentation. However, existing methods typically rely on full fine-tuning, which requires updating all model parameters. Instead of updating the full parameter set, parameter-efficient fine-tuning (PEFT) achieves competitive performance by optimizing only a small subset of parameters. Despite its success, most existing PEFT methods are mainly designed for natural image tasks and fail to account for the unique multiscale characteristics of remote sensing images. To address these challenges, we propose multi-scale cognitive feature refinement (MsRE) tuning, a novel PEFT method tailored for remote sensing semantic segmentation. In particular, MsRE captures multiscale contextual information by applying cognitive operations with different cognitive fields to intermediate features of the backbone. It then introduces a set of learnable tokens to establish interactions with features at different scales, enabling precise feature refinement and progressive feature propagation across network layers. This mechanism enhances the model's ability to understand complex remote sensing scenes and improves downstream segmentation performance. With significantly fewer trainable parameters, MsRE provides an efficient yet effective solution for adapting VFMs to remote sensing segmentation tasks. Extensive experiments demonstrate that MsRE achieves competitive segmentation performance with substantially fewer trainable backbone parameters, providing a favorable balance between accuracy and parameter efficiency. The project is available at http://woldier.top/MsRE**

***Index Terms*—Parameter-efficient fine-tuning (PEFT), remote sensing, semantic segmentation, vision foundation models (VFMs).**

I. Introduction

With the rapid advancement of sensor technology, the number of sensing devices has increased significantly, enabling the large-scale collection of remote sensing images worldwide [1]. The accumulation of remote sensing data, along with the growing prevalence of remote sensing big data technologies, has greatly promoted the development of Earth observation (EO) while simultaneously posing significant challenges to interpretation methods. Semantic segmentation, which assigns a semantic label to each pixel in an image, is one of the most representative approaches for EO interpretation and has been widely applied in land use and land cover mapping [2], [3], [4], environmental monitoring [5], and urban sustainable development [6].

With the rapid development of deep learning, significant progress has been achieved in scene classification [7], [8], [9], object detection [10], [11], [12], and semantic segmentation [13], [14], [15]. In the field of semantic segmentation, fully convolutional networks (FCNs) [16] were first proposed and successfully applied to end-to-end pixelwise classification. These models typically consist of an encoder and a decoder. The encoder is responsible for extracting hierarchical features from the input image while progressively reducing spatial resolution, whereas the decoder upsamples the encoded features to generate dense predictions. This encoder–decoder architecture has since become a fundamental paradigm and has been widely adopted in subsequent studies, leading to many successful models such as DeepLabV3 + [17] and UPerNet [18]. Convolution-based methods are effective at capturing fine-grained local contextual information, which underpins their strong capability in feature representation and pattern recognition. However, despite their strength in modeling local context, convolutional operations are inherently limited by fixed receptive fields, making it difficult to capture global context and long-range dependencies. Vaswani et al. [19] introduced the Transformer, a neural architecture that leverages attention mechanisms to model global context and long-range relationships. Initially developed for natural language processing, the Transformer has achieved remarkable performance and has become a foundational component in modern NLP models. Building upon this success, Dosovitskiy et al. [20] proposed the vision transformer (ViT), which extends the Transformer architecture to computer vision tasks. In ViT, an input image is divided into a sequence of fixed-size patches (e.g., 16×16), which are then processed by the Transformer to capture global relationships and generate global feature representations. With the support of large-scale pretraining datasets,

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ViT achieves highly competitive performance, demonstrating its strong feature extraction capability and great potential in visual understanding. Subsequently, numerous Transformer-based models have been proposed for various vision tasks, particularly semantic segmentation [14], [15], [21], [22], [23]. In the remote sensing domain, many Transformer-based segmentation models have also been developed. For example, Luo et al. [13] proposed FSegNet by combining CNNs and ViTs, while Wang et al. [24] introduced a Transformer-based decoder to construct a U-Net-like architecture. However, these methods typically rely on relatively lightweight backbones, which limit their representation capacity and overall performance.

In recent years, with the advancement of pretraining techniques, large-scale vision foundation models (VFMs), such as MAE [25], DINOv2 [26], DINOv3 [27], SatMAE [28], and ScaleMAE [29], have demonstrated strong robust zero-shot comprehension capabilities when trained on large-scale natural or remote sensing datasets. This has led to their widespread adoption across various tasks. However, VFMs generally contain a large number of parameters; for example, ViT-large has approximately 300M parameters, whereas the backbone of some segmentation models, such as Mix-Transformer [22], contains only about 80M parameters. This substantial increase in model size makes fine-tuning VFMs as segmentation backbones for downstream tasks highly inefficient. Consequently, the traditional "pretrain then fine-tune" paradigm has been evolving. A key challenge in adapting VFMs to downstream tasks lies in efficiently preserving and exploiting their inherent capabilities. In this context, parameter-efficient fine-tuning (PEFT) methods [30], [31], [32] have emerged as a promising solution due to their favorable tradeoff between parameter efficiency and performance. However, existing PEFT methods are primarily designed for natural image tasks. Compared to natural images, remote sensing imagery exhibits significantly larger variations in object scales, which limits the effectiveness of these methods in remote sensing scenarios.

Fig. 1 illustrates the motivation and key differences between existing methods and the proposed approach. Traditional methods either rely on lightweight backbones with limited representation capacity or adopt full fine-tuning of VFMs at the cost of high computational overhead. Although PEFT methods alleviate this issue by updating only a small subset of parameters, they fail to adequately model the complex multiscale structures inherent in remote sensing images. To address these limitations, we propose multi-scale cognitive feature refinement (MsRE) tuning, which enables efficient adaptation of VFMs by incorporating multiscale cognitive feature refinement, achieving a favorable balance between segmentation accuracy and parameter efficiency.

In this article, we propose MsRE tuning, a scale-aware feature-refinement PEFT method tailored for remote sensing semantic segmentation. Unlike conventional PEFT methods that mainly adapt weights, bottleneck modules, or input prompts, MsRE inserts lightweight refinement layers into the frozen VFM backbone to explicitly model multiscale spatial contexts and propagate the refined representations to subsequent blocks. In particular, MsRE applies cognitive operations

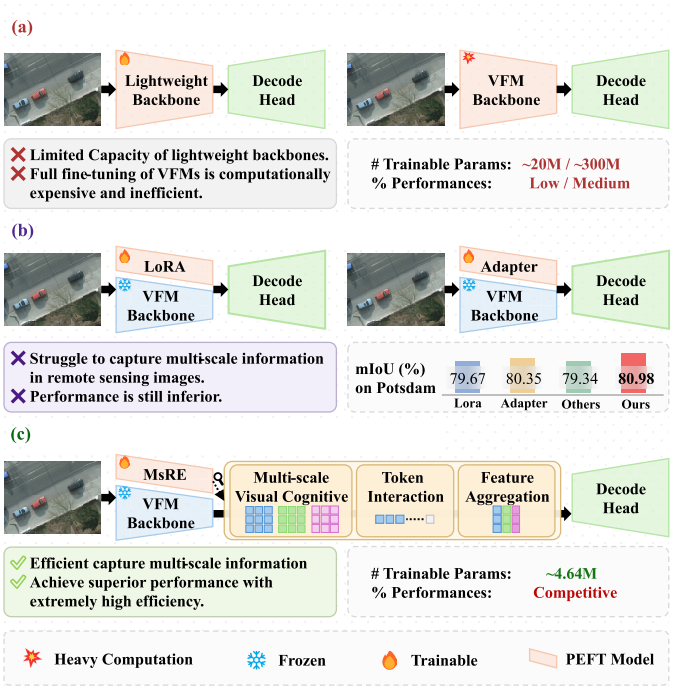


Fig. 1. Comparison of different adaptation paradigms for remote sensing segmentation. (a) Traditional methods show limited performance, while VFM full fine-tuning is inefficient. (b) Existing PEFT methods improve efficiency but fail to capture multiscale characteristics. (c) Proposed MsRE enables efficient and effective adaptation by explicitly refining multiscale features.

with different cognitive fields to intermediate features of the backbone, enabling the extraction of rich multiscale contextual information. It then introduces a set of learnable tokens to establish interactions with features of different scales, thereby enabling precise feature refinement and propagating refined features to subsequent layers of the backbone network. This process enhances the model's ability to understand complex remote sensing characteristics and improves performance on downstream segmentation tasks. We conduct extensive experiments on the Potsdam, Vaihingen, and LoveDA datasets to validate the effectiveness of MsRE. Compared with existing remote sensing segmentation methods and PEFT approaches, MsRE achieves competitive performance while requiring only a small number of trainable parameters. Furthermore, ablation studies are conducted to analyze the effectiveness of MsRE and its generalization across different backbones and decoder heads. The main contributions of this work are summarized as follows.

1) We propose a novel PEFT method, termed MsRE, which is specifically designed for remote sensing semantic segmentation. MsRE effectively adapts VFMs to downstream segmentation tasks by introducing only a small number of trainable parameters.
2) We design a multiscale feature refinement mechanism that leverages convolutional operations with different cognitive fields and token-based interactions to capture scale-aware representations, enabling better modeling of the inherent multiscale characteristics in remote sensing imagery.

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3) We conduct extensive experiments on multiple benchmark datasets, including Potsdam, Vaihingen, and LoveDA. The results demonstrate that MsRE achieves competitive segmentation performance with superior parameter efficiency, especially when compared with existing PEFT methods.

II. Related Work

A. Semantic Segmentation of Remote Sensing Images

Semantic segmentation of remote sensing images is a fundamental task in remote sensing imagery understanding [33], [34]. In recent years, models based on convolutional neural networks (CNNs) and Transformers have significantly improved segmentation performance.

For example, Mou et al. [35] introduced dedicated spatial and channel relationship modeling components to explicitly capture dependencies across spatial locations and feature channels, thereby enhancing contextual interactions. Zhang et al. [1] proposed a hybrid architecture that integrates Transformers with CNNs for semantic segmentation of very high-resolution remote sensing images. Wang et al. [24] developed a Transformer-based decoder and constructed a U-Net-like architecture, termed UNetFormer, for real-time urban scene segmentation. Yue et al. [36] further proposed a lightweight segmentation model that combines the advantages of CNNs and Transformers. However, CNN-based methods inherently struggle to capture long-range dependencies, and rapid upsampling during decoding often leads to undersampling at object boundaries, resulting in inaccurate edge predictions. To address this issue, Huang et al. [37] proposed a multiscale edge optimization network, named MSEONet, to improve boundary quality through explicit edge modeling. In addition, Zhou et al. [38] introduced an edge-guided encoder that augments the Pyramid ViT backbone with a learnable edge-guided feature module, along with a decoder based on a feature-aligned pyramid network enhanced by adaptive spatial feature fusion. Despite their effectiveness, these methods typically rely on relatively lightweight backbones, which constrain their representation capacity and limit segmentation performance. On the contrary, large-scale VFMs exhibit substantially stronger feature extraction capabilities, making them a promising choice for improving segmentation accuracy.

B. Vision Foundation Models

VFMs [39], [40], [41], [42] have achieved remarkable progress and demonstrated strong generalization capabilities across various domains. For instance, MAE [25] leverages masked image modeling to learn latent representations, while DINOv2 [26] achieves strong performance through large-scale self-supervised pretraining. DINOv3 [27] further scales up the training dataset and introduces improved pretraining strategies, resulting in more powerful feature representations. In the remote sensing domain, SatMAE [28] adapts the MAE framework to remote sensing images, and ScaleMAE [29] enhances representation learning by incorporating multiscale masked image modeling. Moreover, DINOv3 [27] has been extended to the remote sensing domain through large-scale data collection and pretraining, further improving its applicability. Although VFMs exhibit strong feature extraction capabilities, fine-tuning is typically required when transferring them to downstream tasks to achieve optimal performance. However, full fine-tuning is computationally expensive. PEFT has therefore emerged as a practical alternative, enabling efficient adaptation of VFMs to downstream tasks with significantly reduced computational overhead.

C. Parameter-Efficient Fine-Tuning

PEFT [14], [30], [31], [43], [44] has emerged as an effective paradigm for adapting large-scale pretrained models to downstream tasks with minimal computational overhead. Instead of updating all model parameters, PEFT methods introduce a small number of learnable parameters while keeping the majority of the backbone frozen, thereby significantly improving training efficiency without sacrificing performance. Among existing approaches, LoRA [30] decomposes weight updates into low-rank matrices and injects them into attention layers, enabling efficient adaptation with negligible parameter overhead. Adapter-based methods insert lightweight bottleneck modules into each layer of the backbone, allowing task-specific information to be learned while preserving the pretrained representations. Building upon this idea, AdaptFormer [32] introduces adapters specifically designed for ViTs, which replace the MLP blocks in Transformer encoders with lightweight adaptation layers to better capture visual features. Yin et al. [45] introduced Mona, which adds an extra layer after the attention and MLP layers to learn visual features. Although these methods have demonstrated strong performance across various vision tasks, they primarily focus on general feature adaptation and do not explicitly consider the multiscale characteristics inherent in remote sensing images. As a result, their ability to handle complex structures and objects with large-scale variations remains limited.

Different from LoRA, Adapter, and AdaptFormer, which mainly perform parameter-efficient adaptation through low-rank weight updates or bottleneck transformations, MsRE focuses on refining intermediate feature representations with explicit scale-aware spatial contexts. Moreover, unlike prompt or token tuning methods that typically append learnable tokens to guide the backbone representation, the learnable tokens in MsRE interact with multiscale feature responses generated by different cognitive fields and are used to produce residual feature refinements. Compared with conventional multiscale segmentation architectures, MsRE does not rely on a fully trainable multiscale encoder–decoder design; instead, it provides a lightweight plug-in refinement mechanism for adapting frozen VFMs to remote sensing segmentation.

III. Method

A. Problem Setup

In the pretraining and fine-tuning framework, the source domain D_s refers to the domain where the model is pretrained, and the target domain refers to the domain D_t where the model is fine-tuned to perform the corresponding downstream task, such as sense classification, change detection, or semantic

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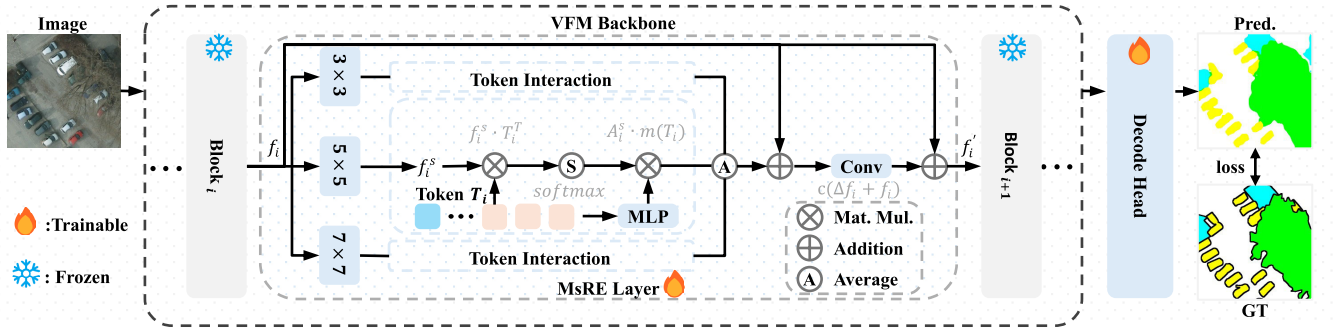


Fig. 2. Overview of MsRE. The proposed MsRE consists of three key parts: multiscale visual cognitive, token interaction, and feature aggregation. The multiscale visual cognitive module extracts features of different scales from f_i using convolutions with varying cognitive fields. The token interaction module further refines these multiscale features by modeling their interactions with learnable tokens. Finally, the feature aggregation module integrates the refined multiscale features and performs the final refinement to produce the output representation.

segmentation. From D_s we obtain a pretrained backbone with parameters θ_s via supervised learning [20], [46] on a large labeled images $D_s = \{x_i, y_i\}_{i=1}^{N_s}$

$$\arg\min_{\theta_s,\delta_s} = \sum_{(x,y)\in D_s} \mathcal{L}_{sl}\left(\mathcal{E}\left(x;\ \theta_s\right) \circ \mathcal{D}\left(\cdot;\ \delta_s\right),\ y\right) \tag{1}$$

where $\mathcal{E}$ denotes backbone with parameters θ_s and $\mathcal{D}$ denotes pretrain head with parameters δ_s, while $\circ$ denotes the function composition operator.

Alternatively, we can obtain backbone with parameters θ_s via self-supervised learning [25], [26], [27] on a large unlabeled images $D_s = \{x_i\}_{i=1}^{N_s}$

$$\arg\min_{\theta_s,\delta_s} = \sum_{x\in D_s} \mathcal{L}_{ssl}\left(\mathcal{E}\left(x;\ \theta_s\right) \circ \mathcal{D}\left(\cdot;\ \delta_s\right)\right). \tag{2}$$

To achieve strong performance on downstream tasks, the pretrained backbone is typically further fine-tuned on the target domain $D_t = \{x_i, y_i\}_{i=1}^{N_t}$, where task-specific supervision is available. The fine-tuning process can be formulated as follows:

$$\arg\min_{\theta_t,\delta_t} = \sum_{(x,y)\in D_t} \mathcal{L}_{task}\left(\mathcal{E}\left(x;\ \theta_t\right) \circ \mathcal{H}\left(\cdot;\ \delta_t\right),\ y\right) \tag{3}$$

where θ_t is initialized from θ_s and $\mathcal{H}$ denotes task-specific head with parameters δ_t. However, full fine-tuning is computationally inefficient, as it requires updating all parameters of the backbone network. This inevitably leads to increased computational and memory costs, hindering the efficient adaptation of the pretrained backbone to downstream tasks.

Instead of updating the full parameter set, PEFT keeps the pretrained backbone θ_s frozen and introduces a small number of learnable parameters $\Delta\theta$, where $\Delta\theta \ll \theta_s$. The optimization objective is

$$\arg\min_{\Delta\theta,\delta_t} = \sum_{(x,y)\in D_t} \mathcal{L}_{task}\left(\mathcal{E}\left(x;\ \theta_s, \Delta\theta\right) \circ \mathcal{H}\left(\cdot;\ \delta_t\right),\ y\right). \tag{4}$$

However, most existing PEFT methods are developed for natural language processing [30], [31] or general vision tasks [32], [43], without considering the unique characteristics of remote sensing imagery. Compared with natural images, remote sensing images exhibit more complex object structures and extreme scale variations, which significantly degrade the performance of these methods in downstream tasks, especially semantic segmentation. To address this issue, we propose MsRE, a PEFT framework tailored for remote sensing semantic segmentation.

B. Overview of MsRE

As illustrated in Fig. 2, the proposed MsRE framework mainly consists of two components: a VFM backbone and a decoder head. Within the backbone, one MsRE layer is inserted after each ViT Transformer block to refine the intermediate features of the VFM. For DINOv3 ViT-Large, this results in $L = 24$ inserted MsRE layers.

Given an input image, it is first transformed into a sequence of tokens through patch embedding and positional encoding, resulting in an input representation f_0. The token sequence is then fed into the backbone for feature extraction, and the extracted features are subsequently passed to the decoder head to generate the final segmentation prediction.

In the backbone, all original blocks remain frozen to preserve the pretrained knowledge of the VFM to the greatest extent. On the contrary, the inserted MsRE layers are learnable and contain only a small number of parameters. These layers refine the output features from the preceding block in an efficient manner. In particular, for each input feature f_i, MsRE first applies convolutions with different cognitive fields to capture multiscale features, denoted as f_i^s. Then, each scale-specific feature undergoes a token interaction module to extract scale-aware discriminative representations. The resulting features are subsequently aggregated via an averaging scheme and combined with the original input feature f_i through a residual connection. Finally, a residual Conv is applied to produce the refined feature representation f_i'.

C. Details of MsRE

Considering a VFM equipped with MsRE layers for feature refinement, given an input image x, the data flow within the backbone can be formulated as follows:

$$f_i' = f_i + MsRE\left(f_i\right), \quad 1 \le i \le L \tag{5}$$

where $f_i = Block_i(f_{i-1}')$, and $f_0' = f_0 = pe(x) + pos(x)$, with $pe(x)$ and $pos(x)$ denotes the *patch embedding* and *positional*

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encoding respectively. In implementation, this refinement is layerwise, i.e., the ith Transformer block is followed by its own refinement module $MsRE_i$, and the refined feature is equivalently written as $f_i' = f_i + MsRE_i(f_i)$.

1) Multiscale Visual Cognitive: In remote sensing images, object scales vary significantly. Therefore, capturing features with different cognitive fields is crucial for modeling multiscale information. To this end, we introduce multiple convolutional operations with different cognitive fields. In particular, the input feature f_i is processed by three visual cognitive convolutions with kernel sizes of 3×3, 5×5, and 7×7, respectively, to obtain multiscale features

$$f_i^s = Conv_s(f_i), \quad s \in \{1, 2, 3\}. \tag{6}$$

Subsequently, features at each scale are further processed via a token interaction module to capture scale-specific representations.

In this work, a cognitive field refers to the spatial support used by an MsRE layer to perceive and refine a reshaped token feature at a specific scale. It is parameterized by the kernel size of the local cognitive operation, such as 3×3, 5×5, and 7×7, but it is not identical to the conventional receptive field. The receptive field describes the accumulated input region that can influence a feature through the whole network, whereas the cognitive field denotes an explicit scale branch within a single MsRE layer for producing scale-aware residual refinement. Thus, the multiscale cognitive operation is not used as a simple multikernel feature aggregation module. Each cognitive-field response is further coupled with learnable token interaction, enabling the model to refine scale-specific features before they are aggregated and propagated through the frozen backbone.

2) Token Interaction: To better establish interactions between tokens and multiscale features, we introduce a set of learnable tokens

$$T = \left\{ T_i \in \mathbb{R}^{m \times c} \middle| 1 \le i \le L \right\} \tag{7}$$

where m denotes the token length. These tokens are randomly initialized and are used to interact with features at different scales. The default token length is set to $m = 100$ according to the token-length ablation. Each MsRE layer has its own learnable token set T_i, and tokens are not shared across different Transformer blocks. Within the same MsRE layer, the same token set T_i is shared across different scale branches generated by different cognitive fields, which encourages cross-scale interaction while keeping the parameter cost low. Inspired by [19] and [20], the token interaction module computes the similarity between multiscale features f_i^s and tokens T_i as follows:

$$A_i^s = Softmax\left(\frac{f_i^s \odot T_i^T}{\sqrt{c}} \right), \quad A_i^m \in \mathbb{R}^{n \times m}. \tag{8}$$

This operation captures the relationships between multiscale features and tokens, facilitating effective multiscale feature modeling. The output of the token interaction is given by

$$\Delta f_i^s = A_i^s \odot mlp(T_i). \tag{9}$$

TABLE I
OVERVIEW OF THE POTSDAM, VAIHINGEN, AND LOVEDA DATASETS

Dataset	Resolution	Crop / Stride	Train split	Valid split
Potsdam [47]	6000×6000	512 / 512	24 (2904◇)	14 (1694◇)
Vaihingen [47]	2906×2273	512 / 256	16 (1296◇)	17 (440◇)
LoveDA [48]	1024×1024	-	2522	1671

◇ denotes the number of images after cropping. The image resolution vary within the Vaihingen dataset, hence the values provided represent averages.

3) Feature Aggregation: We aggregate the outputs from different scales by averaging the refined features Δf_i^s

$$\begin{aligned} \bar{f}_i &= f_i + avg\left(\Delta f_i^1, \Delta f_i^2, \Delta f_i^3\right) \\ f_i' &= f_i + conv\left(\bar{f}_i\right) \end{aligned} \tag{10}$$

where a 1×1 convolution is applied to further refine the aggregated features, and residual connections are employed in both steps to preserve the original representations.

This process progressively aligns the learned features with the multiscale characteristics of remote sensing images, thereby enhancing the model's visual understanding capability.

D. Optimization Objective

For the semantic segmentation task, the goal is to assign a semantic label to each pixel of the input image. Given a target dataset $D_t = \{x_i, y_i\}_{i=1}^{N_t}$, where y denotes the pixelwise annotation, the model is optimized to minimize the segmentation loss over all pixels. Following the PEFT paradigm, we keep the pretrained backbone parameters θ_s frozen and only optimize the parameters of the MsRE layers $\Delta\theta$ together with the decoder head δ_t. The overall optimization objective is defined as follows:

$$\arg\min_{\Delta\theta, \delta_t} = \sum_{(x,y) \in D_t} \mathcal{L}_{seg}\left(\mathcal{E}(x;\ \theta_s, \Delta\theta) \circ \mathcal{H}(\cdot;\ \delta_t), y\right) \tag{11}$$

where $\mathcal{E}(\cdot;\ \theta_s, \Delta\theta)$ denotes the backbone with inserted MsRE layers, and $\mathcal{H}(\cdot;\ \delta_t)$ represents the decoder head for dense prediction.

In this work, $\mathcal{L}_{seg}$ is defined as the pixelwise cross-entropy loss, which measures the discrepancy between the predicted segmentation map and the ground-truth labels. By optimizing this objective, MsRE enables effective adaptation of the pretrained backbone to the semantic segmentation task while maintaining high parameter efficiency.

IV. EXPERIMENTAL

A. Experimental Setup

1) Datasets: As illustrated in Table I, this article focuses on evaluating semantic segmentation performance on the Potsdam,[1] Vaihingen,[2] and LoveDA[3] datasets.

[1]https://www.isprs.org/resources/datasets/benchmarks/UrbanSemLab/2d-sem-label-potsdam.aspx

[2]https://www.isprs.org/resources/datasets/benchmarks/UrbanSemLab/2d-sem-label-vaihingen.aspx

[3]https://github.com/Junjue-Wang/LoveDA

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TABLE II
SEMANTIC SEGMENTATION RESULT (MIOU%) COMPARISONS ACROSS DIFFERENT METHODS ON THE POTSDAM, VAIHINGEN, AND LOVEDA DATASETS

Method	Pre-Train Method	Pre-Train Dataset	Backbone and Scale	Fine-Tune Params ↓	Potsdam (mIoU%)	Vaihingen (mIoU%)	LoveDA (mIoU%)	Average mIoU% ↑
I. Specific Segmentation Methods								
1. General Segmentation Methods								
DeepLabV3+ [17]	Supervised	ImageNet1K	Resnet-50	25.2 M	73.43	71.87	45.33	63.54
UperNet [18]	Supervised	ImageNet1K	Resnet-50	25.2 M	73.43	71.86	46.28	63.85
Segformer [22]	Supervised	ImageNet1K	MiT-B5	89.9 M	79.01	74.38	49.69	67.69
Segmenter [23]	Supervised	ImageNet1K	ViT-Large	304.3 M	78.24	73.12	53.34	68.23
2. Remote Sensing Segmentation Methods								
UNetFormer [24]	Supervised	ImageNet1K	ResNet-18	11.6 M	74.16	72.38	46.83	64.45
FSegNet [13]	Supervised	ImageNet1K	FasterViT-3	159.5 M	78.17	74.51	48.93	67.20
MSEONet [37]	Supervised	ImageNet1K	Resnet-101	44.5 M	74.51	70.08	45.17	63.25
EMGSNet [38]	Supervised	ImageNet1K	PVT-Base	42.1 M	79.64	75.84	52.95	69.47
II. Vision Foundation Models								
Full Fine-Tuning	DINOv3 [27]	LVD-1689M	ViT-Large	303.1 M	**81.04**	75.12	54.61	70.25
Frozen	DINOv3 [27]	LVD-1689M	ViT-Large	0 M	79.21	74.34	52.55	68.70
i. General PEFT Methods								
LoRA [30]	DINOv3 [27]	LVD-1689M	ViT-Large	9.43 M	79.67	73.25	53.19	68.70
Adapter [31]	DINOv3 [27]	LVD-1689M	ViT-Large	6.24 M	80.35	73.91	54.18	69.48
AdaptFormer [32]	DINOv3 [27]	LVD-1689M	ViT-Large	3.12 M	79.34	76.12	15.18	56.88
Mona [45]	DINOv3 [27]	LVD-1689M	ViT-Large	5.08 M	80.39	74.96	53.87	69.74
ii. Remote Sensing PEFT Methods								
AiRs [49]	DINOv3 [27]	LVD-1689M	ViT-Large	4.24 M	79.94	74.78	54.11	69.61
Tea [50]	DINOv3 [27]	LVD-1689M	ViT-Large	11.32 M	80.43	75.01	54.36	69.93
iii. Ours Approach								
MsRE (ours)	DINOv3 [27]	LVD-1689M	ViT-Large	4.64 M	80.98	**76.72**	**54.90**	**70.86**

The results in each column is highlighted in **bold**. All Vision Foundation models are evaluated using the Segmenter head [23] for fair comparison. "Fine-Tune Params" refer to the number of trainable parameters in the backbone on the downstream tasks. "Supervised" stands for supervised learning.

a) Potsdam: The Potsdam [47] dataset consists of 38 high-resolution remote sensing images, each with a spatial resolution of 6000 × 6000 pixels and a ground sample distance (GSD) of 5 cm. The dataset contains six semantic classes, including *impervious surface, building, low vegetation, tree, car,* and *clutter*. It provides three image modalities: RGB, IRRG, and RGBIR. In this work, we adopt the RGB modality. To reduce computational overhead, all images are cropped into 512 × 512 patches using a sliding window with a stride of 512, resulting in a total of 4598 image patches. These patches are further split into 2904 training samples and 1694 testing samples.

b) Vaihingen: The Vaihingen [47] dataset contains 33 high-resolution remote sensing images, with spatial resolutions ranging from 1996 × 1996 to 3816 × 2550, and a GSD of 9 cm. Similar to the Potsdam dataset, it includes six classes: *impervious surface, building, low vegetation, tree, car,* and *clutter*. The dataset provides only the IRRG modality. In our experiments, images are cropped into 512 × 512 patches using a sliding window with a stride of 256, resulting in 1696 image patches. These are further divided into 1296 training samples and 440 testing samples.

c) Loveda: The LoveDA [48] dataset consists of 5987 high-resolution remote sensing images collected from Nanjing, Guangzhou, and Wuhan, each with a resolution of 1024 × 1024. The dataset contains seven semantic classes, including *agriculture, forest, barren, water, road, building* and *background*. It contains two distinct domains, namely *urban* and *rural*. In particular, the urban domain includes 1883 images, which are split into 1156 training samples and 677 validation samples. The rural domain contains 2358 images, with 1366 training samples and 992 validation samples. In this work, data from both domains are combined for training and evaluation.

2) Implementation Details: The implementation is based on PyTorch and the MMSegmentation[4] framework. We adopt DINOv3 [27] ViT-Large pretrained on LVD-1689M as the backbone in this work. Considering the strong compatibility between Segmenter [23] and ViT, we follow prior work [14] and employ Segmenter as the decoder head.

During training, the AdamW optimizer is used with an initial learning rate of 1×10^{-4}. To further ensure training stability, we adopt a poly learning rate schedule with a linear warmup of 1500 iterations. The batch size is set to 2, and the model is trained for 40k iterations. All experiments are conducted on an NVIDIA RTX 4090D GPU. Unless otherwise specified, all datasets use the same MsRE hyperparameters, including the token length $m = 100$, the cognitive fields $\{3 \times 3, 5 \times 5, 7 \times 7\}$, and the insertion strategy after every Transformer block. We do not tune these hyperparameters separately for Potsdam, Vaihingen, or LoveDA.

B. Performance Comparison

As illustrated in Table II, this part compares several mainstream semantic segmentation methods, including DeepLabV3 + [17], UperNet [18], SegFormer [22], and Segmenter [23]. In addition, several remote sensing-specific segmentation methods are included for comparison, such as UNetFormer [24], FSegNet [13], MSEONet [37], and EMGSNet [38]. To further evaluate the proposed method,

[4]https://github.com/open-mmlab/mmsegmentation

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TABLE III
STABILITY ANALYSIS ON POTSDAM UNDER THREE INDEPENDENT RUNS

Method	Fine-Tune Params	Potsdam mIoU%
Full Fine-Tuning	303.1 M	81.01 ± 0.027
LoRA [30]	9.43 M	79.65 ± 0.026
Tea [50]	11.32 M	80.44 ± 0.015
MsRE (ours)	4.64 M	80.99 ± 0.011

All results are reported as mean ± standard deviation over three independent runs under the same experimental setting. "Fine-Tune Params" refer to the number of trainable parameters in the backbone.

we also compare it with representative PEFT approaches, including LoRA [30], Adapter [31], AdaptFormer [32], and Mona [45]. Moreover, PEFT methods specifically designed for remote sensing are also considered, including AiRs [49] and Tea [50]. It is worth noting that, for a fair comparison, all PEFT methods and the proposed approach adopt the same backbone and decoder head.

From the experimental results, as illustrated in Table II, it can be observed that for specific segmentation methods, the number of backbone parameters is positively correlated with the final prediction accuracy. This is because larger backbones with more parameters generally provide stronger representation capacity and better image understanding. For instance, from DeepLabV3 + to Segmenter, the number of parameters increases by nearly 12 times, resulting in a significant performance improvement. However, larger models typically require substantially more computational resources. Compared with full fine-tuning, PEFT approaches require updating only a small fraction of backbone parameters, making them a more efficient adaptation alternative. As summarized in Table II, MsRE achieves 80.98% mIoU on Potsdam, which is comparable to DINOv3 full fine-tuning (81.04%) while using only 4.64M trainable backbone parameters instead of 303.1M. Compared with LoRA and Tea, MsRE obtains higher mIoU, suggesting that multiscale feature refinement improves PEFT adaptation for remote sensing segmentation. Therefore, MsRE should be understood as a parameter-efficient method that achieves competitive accuracy rather than as a method that universally outperforms full fine-tuning.

To further assess statistical stability and clarify the comparison with full fine-tuning, we report the mean and standard deviation over three independent Potsdam runs listed in Table III. MsRE achieves 80.99% mean mIoU with 4.64M trainable backbone parameters, which is comparable to full fine-tuning (81.01%) while using substantially fewer trainable parameters. The gain over Tea is modest, but the repeated-run statistics provide a more complete basis for interpreting MsRE as a stable and competitive PEFT alternative under the same evaluation setting.

C. Ablation Studies

1) Classwise IoU Comparison: In this section, we present the detailed classwise performance on three datasets, including Potsdam, Vaihingen, and LoveDA, comparing FSegNet [13], Frozen, LoRA [30], and the proposed MsRE. All results are reported in Table IV.

TABLE IV
CLASSWISE IOU COMPARISON ON POTSDAM, VAIHINGEN, AND LOVEDA

Methods	Clut.	Car	Tree	Vege.	Build.	Imp.	mIoU
Class-wise IoU on Potsdam							
FSegNet [13]	41.9	**90.5**	79.4	77.3	93.0	86.6	78.1
Frozen⋆	46.3	89.9	79.9	78.2	93.6	87.0	79.2
LoRA⋆ [30]	48.2	90.0	80.1	78.4	93.7	87.2	79.6
MsRE⋆ (ours)	**51.3**	90.4	**81.2**	**79.8**	**94.5**	**88.4**	**80.9**
Class-wise IoU on Vaihingen							
FSegNet [13]	34.6	**76.6**	83.4	73.8	91.1	**85.6**	74.2
Frozen⋆	46.4	67.9	82.7	73.0	92.0	84.0	74.3
LoRA⋆ [30]	34.8	70.8	83.2	73.3	92.4	83.8	73.2
MsRE⋆ (ours)	**49.6**	70.4	**84.6**	**75.9**	**94.2**	85.4	**76.7**

For class-wise IoU%, only results rounded to one decimal place are reported. Mark ⋆ indicates the method are based on the frozen DINOv3-Large [27]. The abbreviations "Clut.", "Vege.", "Build.", and "Imp." in the table stand for clutter, low vegetation, building, and impervious surface, respectively.

Methods	Argi.	Fore.	Barr.	Water	Road	Build.	Back.	mIoU
Class-wise IoU on LoveDA								
FSegNet [13]	60.6	41.4	14.6	67.5	52.6	61.1	44.3	48.9
Frozen⋆	54.3	43.1	33.9	68.0	49.0	64.4	**54.8**	52.5
LoRA⋆ [30]	**59.2**	43.2	31.7	63.0	53.5	**67.9**	53.5	53.1
MsRE⋆ (ours)	57.4	**47.2**	**32.1**	**72.2**	**53.9**	67.2	54.0	**54.9**

The abbreviations "Agri.", "Fore.", "Barr.", "Build.", and "Back." in the table stand for agriculture, forest, barren, building, and background, respectively.

In Potsdam, it can be observed that FSegNet achieves the best performance on the *car* category, which can be attributed to its multiscale decoder head that is capable of better distinguishing objects with varying shapes. Compared to Frozen, FSegNet shows notable improvement on the *car* category. However, the performance gap between FSegNet and MsRE is marginal, indicating that although MsRE performs segmentation prediction using only the final-layer features, it can still accurately identify cars. This demonstrates that MsRE is capable of extracting more informative and discriminative features. Furthermore, despite the limited number of samples in the *clutter* category, MsRE significantly outperforms other methods, suggesting its stronger generalization ability.

On Vaihingen and FSegNet show competitive performance on the *car* and *impervious surface* categories, particularly achieving an IoU of 85.6 on *impervious surface*. On the contrary, MsRE exhibits superior performance on the *tree* and *low vegetation* categories. Although these categories involve significant variations in shape and scale, MsRE maintains high recognition accuracy. Moreover, MsRE consistently outperforms LoRA on *tree*, *low vegetation*, and *building*, indicating that MsRE can more effectively leverage the inherent multiscale characteristics of remote sensing images to improve prediction performance.

On LoveDA, although MsRE does not show prominent advantages on the *agriculture* and *building* categories, its performance remains comparable to LoRA. On the contrary, MsRE achieves significant improvements on categories with substantial scale variations, such as *barren*, *water*, and *road*.

2) Prediction Visualization Analysis: Fig. 3 presents the visualization results of Frozen DINOv3 [27], LoRA [30], and MsRE on the Potsdam and LoveDA datasets. From the

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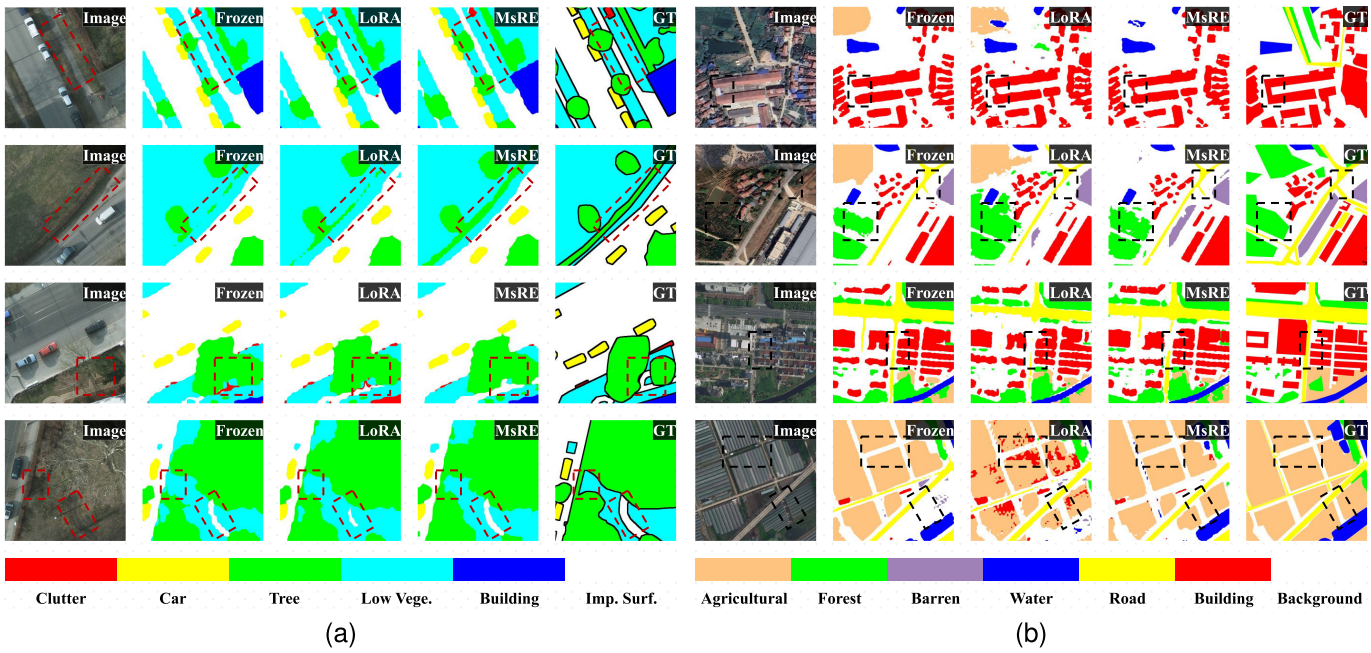


Fig. 3. Visualization of results on (a) Potsdam and (b) LoveDA. Note that all methods are based on DINOv3.

qualitative comparisons, it can be observed that when the backbone is frozen, the model is able to roughly distinguish objects. However, it produces noticeable errors along fine-grained boundaries and detailed structures. This indicates that the pretrained DINOv3 backbone possesses a certain level of feature extraction capability. Nevertheless, the inability to adapt its parameters to specific datasets limits its performance in segmentation tasks. On the contrary, PEFT methods, such as LoRA and MsRE, significantly enhance prediction quality by introducing only a small number of trainable parameters. This demonstrates that PEFT provides an effective balance between training efficiency and segmentation accuracy. However, since LoRA does not explicitly model the inherent multiscale characteristics of remote sensing images, its performance remains limited, particularly in complex regions. Compared with LoRA, the proposed MsRE achieves more accurate and refined predictions. By performing multiscale refinement on intermediate features, MsRE is able to better delineate object boundaries and substantially improve accuracy in challenging regions, further validating its effectiveness in capturing the intrinsic multiscale structures of remote sensing images.

3) Feature Visualization Analysis: We conduct PCA-based feature visualization shown in Fig. 4 to analyze the representation learning behavior of MsRE. In particular, we visualize the intermediate feature representations from Layer 6, 12, 18, and 24 on the Potsdam dataset. As illustrated in the figure, the features exhibit progressively improved intraclass compactness and interclass separability as the network depth increases, which validates the effectiveness of the proposed multiscale cognitive feature refinement.

Furthermore, we observe a clear evolution of semantic focus across different layers. The shallow layers (e.g., Layer 6) primarily capture large-scale structures, such as *building* and *impervious surface*, which are characterized by relatively large spatial extents. As the features propagate to deeper layers, the model gradually shifts its attention toward smaller and more fine-grained objects, such as *car*. In the deepest layer (Layer 24), the features achieve clear separation among all semantic categories, indicating that MsRE effectively integrates multiscale information and progressively refines feature representations to enhance discriminability.

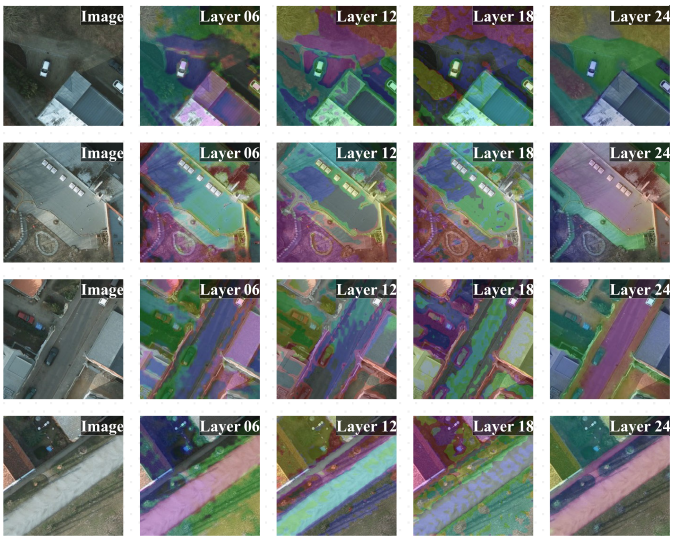


Fig. 4. PCA visualization of feature representations from different layers (Layer 6, 12, 18, and 24) of MsRE on the Potsdam dataset.

4) Performance on Different Backbones: In addition to using DINOv3 [27] as the backbone, we further investigate the adaptability of the proposed MsRE to backbones pretrained with different strategies. In particular, we select representative methods pretrained on natural images, including MAE [25], DINOv2 [26], and DINOv3 [27]. In addition, we consider several representative methods pretrained on remote sensing

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TABLE V
ABLATION STUDIES OF MSRE ON VARIOUS DIFFERENT BACKBONES

Backbone	Pre-Train Dataset	+PEFT Method	Fine-Tune Params †	mIoU%
Natural Images Vision Foundation Models				
MAE [25] (ViT-Large)	ImageNet1K	Frozen	0 M	68.13
		+LoRA	9.43 M	70.03
		+MsRE	4.64 M	**72.19**
DINOv2 [26] (ViT-Large)	LVD-142M	Frozen	0 M	75.79
		+LoRA	9.43 M	76.67
		+MsRE	4.64 M	**78.23**
DINOv3 [27] (ViT-Large)	LVD-1689M	Frozen	0 M	79.21
		+LoRA	9.43 M	79.67
		+MsRE	4.64 M	**80.98**
Remote Sensing Vision Foundation Models				
SatMAE [28] (ViT-Large)	fMoW-RGB	Frozen	0 M	64.85
		+LoRA	9.43 M	64.98
		+MsRE	4.64 M	**67.47**
ScaleMAE [29] (ViT-Large)	fMoW-RGB	Frozen	0 M	68.51
		+LoRA	9.43 M	69.21
		+MsRE	4.64 M	**70.43**
DINOv3 [27] (ViT-Large)	SAT-493M	Frozen	0 M	75.85
		+LoRA	9.43 M	76.47
		+MsRE	4.64 M	**76.52**

Semantic segmentation results are conducted on Potsdam dataset. Mark† denotes the number of trainable parameters in the backbone.

TABLE VI
ABLATION STUDIES OF MSRE ACROSS DIFFERENT BACKBONE SCALES

Pre-Train Method	Pre-Train Params ↓	+PEFT Method	Trainable Params ↓	mIoU%
DINOv3 [27] (ViT-Small)	21.59 M	Frozen	0 M	75.29
		+LoRA	1.77 M	75.83
		+MsRE	0.79 M	**77.49**
DINOv3 [27] (ViT-Base)	85.66 M	Frozen	0 M	76.92
		+LoRA	3.53 M	77.45
		+MsRE	2.17 M	**78.88**
DINOv3 [27] (ViT-Large)	303.1 M	Frozen	0 M	79.21
		+LoRA	9.43 M	79.67
		+MsRE	4.64 M	**80.98**

Semantic segmentation results are reported on the Potsdam dataset. Note that all DINOv3 backbones are pre-trained on the LVD-1689M [27]. Mark† denotes the number of trainable parameters in the backbone during fine-tuning phase.

TABLE VII
ABLATION STUDY OF GENERALIZATION ACROSS DECODER HEADS

Backbone	Decode Head	Decode Params	Selected Feature	mIoU%
DINOv3 [27] +LoRA [30]	Segmenter [23]	28.3 M	[24]	79.67 (+0.00)
	UperNet [18]	121.1 M	[6, 12, 18, 24]	80.25 (+0.58)
	Segformer [22]	13.90 M	[6, 12, 18, 24]	**80.68 (+1.01)**
DINOv3 [27] +MsRE	Segmenter [23]	28.3 M	[24]	80.98 (+0.00)
	UperNet [18]	121.1 M	[6, 12, 18, 24]	**81.47 (+0.49)**
	Segformer [22]	13.90 M	[6, 12, 18, 24]	81.31 (+0.33)

Semantic segmentation results are reported on the Potsdam dataset. All methods use DINOv3 (ViT-Large), pretrained on LVD-1689M, as the backbone. The last column shows mIoU% and improvements over the Segmenter head.

images, including SatMAE [28], ScaleMAE [29], and DINOv3 [27] pretrained on remote sensing data.

The results are presented in Table V. From the experimental analysis, it can be observed that MsRE consistently achieves strong performance across different backbones and outperforms the comparison method LoRA. However, the inherent feature extraction capabilities of different pretrained backbones vary significantly. For example, MsRE achieves only 72.19 mIoU with the MAE backbone, whereas it reaches 80.98 mIoU when using the DINOv3 backbone.

Furthermore, we observe that backbones pretrained on remote sensing data generally underperform those pretrained on natural images. This may be attributed to the fact that pretraining on natural images enables the backbone to learn more generalizable representations, whereas pretraining on remote sensing images tends to yield weaker generalization capability.

5) Ablation Studies on Different Backbone Scales: As illustrated in Table VI, we evaluate the effectiveness of MsRE across different backbone scales, including ViT-Small, ViT-Base, and ViT-Large pretrained from DINOv3 [27]. From the results, two key observations can be drawn. First, MsRE consistently improves segmentation performance across all backbone scales. Compared with LoRA, MsRE achieves higher mIoU while introducing significantly fewer trainable parameters. For example, on ViT-Large, MsRE improves the performance from 79.67% to 80.98% while reducing the number of trainable parameters from 9.43M to 4.64M. This demonstrates that MsRE provides a more favorable parameter and performance tradeoff. Second, the effectiveness of MsRE remains stable as the backbone scale increases. Unlike some PEFT methods whose gains diminish with different models, MsRE maintains consistent improvements from ViT-Small to ViT-Large. This suggests that the proposed multiscale cognitive feature refinement mechanism is complementary to the backbone capacity and can effectively enhance feature representation regardless of model scale. These results verify that MsRE is a scalable and parameter-efficient tuning strategy, capable of consistently improving performance across different backbone sizes.

6) Generalization Across Decoder Heads: For fair comparison, we follow prior works [14], [23] and adopt the Segmenter head [23] in all main experiments. Notably, as illustrated in Table VII, when replacing the decoder with a SegFormer head, our method achieves further improvements, demonstrating its strong compatibility and generalization ability across different segmentation architectures. Compared with LoRA, MsRE consistently achieves higher mIoU under the same decoder head. For instance, when using SegFormer as the decoder head, MsRE achieves 81.31 mIoU, outperforming LoRA, which obtains 80.68. Although LoRA benefits from replacing the Segmenter head with the SegFormer head, yielding a performance gain of 1.01, its performance remains inferior to MsRE with the Segmenter head. This indicates that MsRE is a more general and effective tuning strategy, capable of delivering superior performance across different decoder heads. Furthermore, we observe that replacing the Segmenter head with

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TABLE VIII
STUDY ON DIFFERENT COGNITIVE FIELDS AND AGGREGATE STRATEGY

	Field 3×3	Field 5×5	Field 7×7	Aggregated	mIoU%
1	✓	-	-	Average	80.59
2	-	✓	-	Average	80.63
3	-	-	✓	Average	80.55
4	✓	✓	✓	Weighted	80.95
5	✓	✓	✓	Average	**80.98**

Semantic segmentation results are reported on the Potsdam dataset. All experiments use DINOv3 ViT-Large as the backbone, which is pretrained on LVD-1689M [27].

TABLE IX
EVALUATING THE CONTRIBUTION OF EACH COMPONENT IN THE MSRE

	M.S. Visual Cognitive	Token Interaction	mIoU%
1	✓	-	80.71
2	-	✓	80.56
3	✓	✓	**80.98**

Semantic segmentation results are reported on the Potsdam dataset.

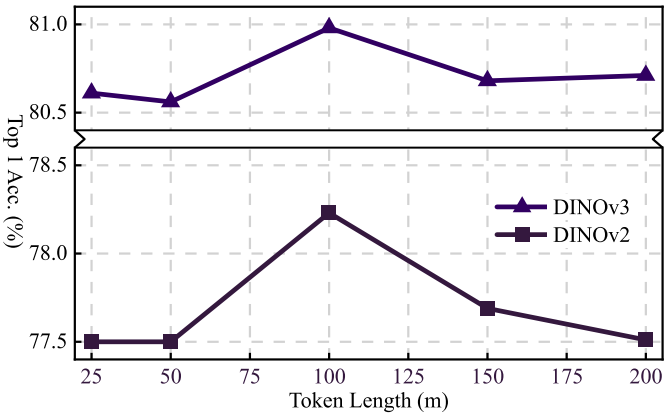


Fig. 5. Ablation study on the impact of token length m. All experiments are conducted using a ViT-Large backbone pretrained with DINOv2 or DINOv3.

TABLE X
ABLATION STUDY ON TOKEN SHARING STRATEGY

Token Strategy	Trainable Params	mIoU (%)
Scale-Specific Tokens	9.47 M	80.57
Across Scales (MsRE)	**4.64 M**	**80.98**

Semantic segmentation results on the Potsdam dataset using DINOv3 (ViT-Large) as the backbone. All token sets are layer-specific and are not shared across different Transformer blocks.

SegFormer leads to consistent performance improvements, despite the reduction in decoder parameters. This improvement can be attributed to the utilization of multilevel intermediate features from the backbone (i.e., [6, 12, 18, 24]), which provide richer multiscale representations. This observation further highlights the importance of leveraging intermediate features for enhancing segmentation accuracy, and also aligns well with the design motivation of MsRE, which explicitly refines features across multiple scales.

7) Component Effectiveness Ablation: In this section, we conducted an ablation study of the MsRE design.

a) Ablation on different cognitive fields: Table VIII investigates the impact of different cognitive fields on model performance. From the experimental results, it can be observed that using a single cognitive field yields relatively limited performance. In particular, configurations with only 3×3, 5×5, or 7×7 fields (Rows 1–3) achieve comparable mIoU scores, indicating that single-scale features are insufficient for effectively capturing the complex structures in remote sensing images. On the contrary, combining multiple cognitive fields enables the model to better exploit multiscale information, resulting in improved segmentation performance.

b) Ablation on feature aggregation: In this work, multiscale features are aggregated via simple averaging. To further analyze the effect of the aggregation strategy, Table VIII also illustrates the comparison of it with a learnable weighted aggregation scheme (Rows 4-5). The results show that different aggregation strategies lead to only marginal performance differences. This suggests that the primary performance gain stems from multiscale feature extraction rather than the specific design of the aggregation mechanism. This observation further validates the effectiveness of MsRE in refining multiscale representations.

c) Ablation on different components: In the proposed MsRE, the multiscale visual cognitive module and the token interaction module are designed for multiscale feature extraction and feature refinement, respectively. To evaluate the effectiveness of these two components, we design two ablation settings. In the first setting, only the multiscale visual cognitive module is retained, and the feature aggregation is directly performed on f_i^s without token interaction. In the second setting, only the token interaction module is used, where a single-scale feature f_i is adopted as input. The results are reported in Table IX. It can be observed that using only the multiscale visual cognitive module leads to competitive performance, demonstrating the benefit of multiscale feature extraction. However, due to the absence of token interaction for further refinement, the overall performance remains limited. On the contrary, when only the token interaction module is applied without multiscale features, the performance degrades noticeably. This indicates that relying solely on single-scale features is insufficient for capturing the complex structures in remote sensing images. These observations demonstrate that multiscale feature extraction and token-based refinement are complementary. Only by effectively integrating both components can the model achieve superior segmentation performance.

d) Ablation on learnable token length: In this work, MsRE employs learnable tokens $T_i \in \mathbb{R}^{m \times c}$ to further refine multiscale features. We investigate the effect of different token lengths m, ranging from 25 to 200. The experimental results are shown in Fig. 5. It can be observed that the model achieves the best performance when the token length is set to 100. Therefore, we adopt $m = 100$ as the optimal hyperparameter in this work.

e) Ablation on token sharing strategy: In the proposed MsRE, a unified set of learnable tokens T_i is shared across

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TABLE XI

ABLATION STUDY ON MSRE INSERTION POSITIONS ON POTSDAM

Strategy	Blocks	Params	mIoU(%)
Early	[1–8]	2.84 M	80.22
Middle	[9–16]	2.84 M	80.36
Late	[17–24]	2.84 M	80.61
All (MsRE)	[1–24]	4.64 M	**80.98**

All experiments use DINOv3 ViT-Large as the backbone. "All" denotes the default setting where one MsRE layer is inserted after each Transformer block.

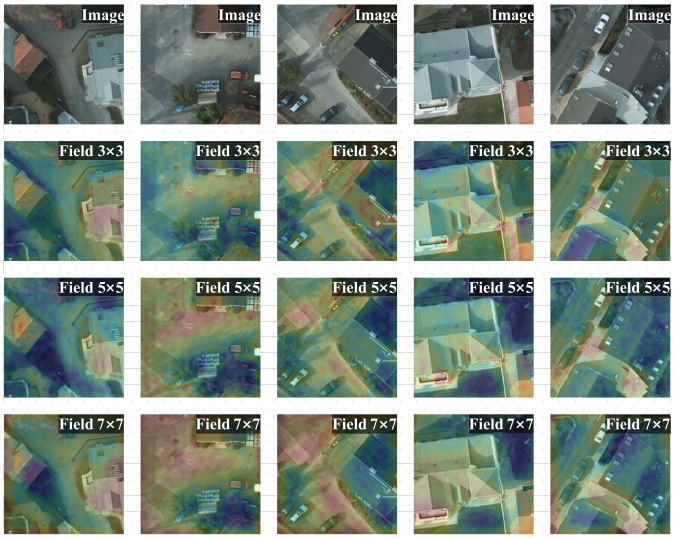


Fig. 6. Visualization of multiscale feature-token similarity with different cognitive fields (3×3, 5×5, and 7×7). All experiments use DINOv3-Large as the backbone.

different scales to refine multiscale features. To further investigate the impact of the token design, we conduct an ablation study by assigning independent tokens for each scale, i.e., using different token sets for features extracted from different cognitive fields. As illustrated in Table X, the experimental results indicate that using separate tokens for each scale leads to inferior performance compared to the shared-token design. This indicates that sharing a unified set of tokens across multiple scales is more effective for feature refinement. A possible explanation is that shared tokens can capture cross-scale semantic consistency and facilitate information exchange among features of different resolutions. On the contrary, using independent tokens restricts the interaction within each scale, limiting the model's ability to learn unified and discriminative representations. It should be noted that the token sharing in Table X refers to sharing tokens across scale branches within the same MsRE layer. Tokens are still layer-specific and are not shared across different Transformer blocks. We further analyze different insertion positions summarized in Table XI. The early, middle, and late settings insert MsRE layers only after blocks [1–8], [9–16], and [17–24], respectively, while the all-block setting inserts one MsRE layer after every Transformer block. Compared with partial insertion, the all-block strategy enables progressive scale-aware feature refinement throughout the backbone, and is therefore adopted as the default setting.

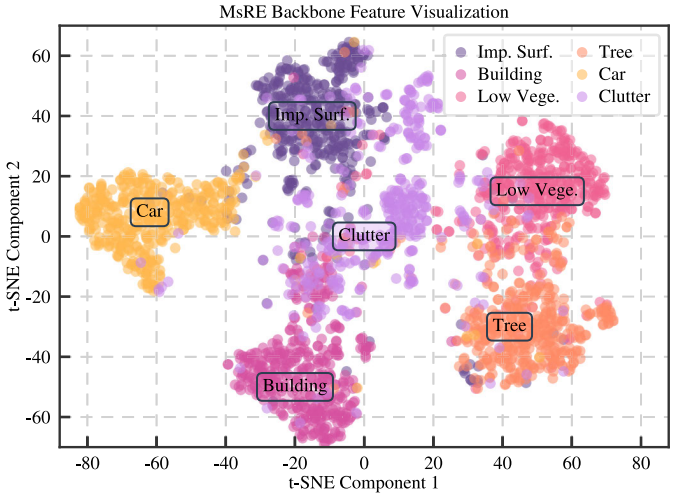


Fig. 7. t-SNE embedding of the last layer of the backbone on Potsdam. The abbreviations "Imp. Surf." and "Low Vege." in the figure stand for impervious surface and low vegetation, respectively.

8) Learnable Tokens Visualization: Fig. 6 visualizes the attention between learnable tokens and multiscale features. It can be observed that features extracted with smaller cognitive fields tend to focus on objects with smaller spatial scales, while larger cognitive fields gradually shift attention toward larger objects. This indicates that the learnable tokens effectively adapt to features at different scales and attend to objects of corresponding sizes. These results further demonstrate the capability of MsRE to capture scale-aware representations through token-based interactions.

9) t-SNE Visualization: We further analyze the feature representations using t-SNE, as shown in Fig. 7. It can be observed that features of different categories form well-separated clusters in the embedding space, indicating strong interclass separability. In particular, categories such as *car* and *building* exhibit highly compact distributions, demonstrating that the proposed method learns discriminative and robust representations. Moreover, MsRE effectively distinguishes objects with different spatial scales. For instance, small objects such as *car* are clearly isolated from other categories, while large-scale structures such as *building* and *impervious surface* also form stable clusters. This validates the effectiveness of the proposed multiscale cognitive feature refinement in capturing scale-aware information. We also observe that certain challenging categories, such as *clutter*, show partial overlap with other classes. This is expected due to their inherent ambiguity and high interclass similarity in remote sensing imagery. Nevertheless, the overall distribution remains well-structured, indicating that MsRE significantly enhances the quality of feature representations. The visualization result demonstrates that MsRE produces more discriminative and structured feature embeddings, which contribute to improved segmentation performance.

10) Visualization of Grad-CAM: We further analyze the model behavior using Grad-CAM visualizations, as shown in Fig. 8. The activation maps reveal that MsRE is able to focus accurately on object regions. In particular, MsRE produces

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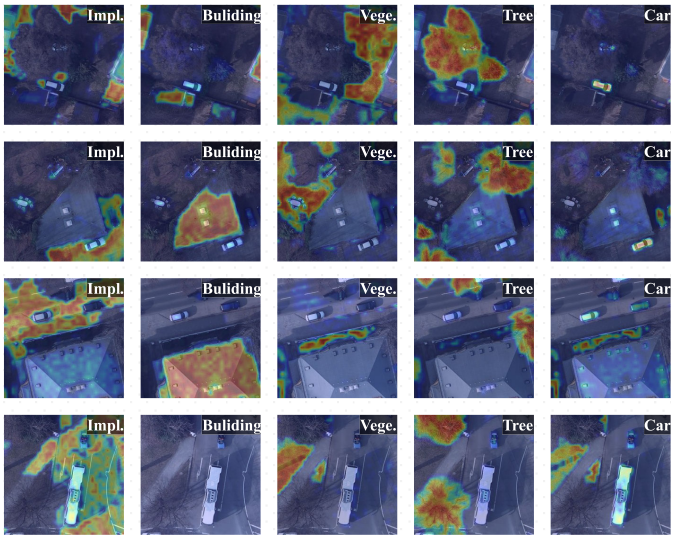


Fig. 8. Visualization of Grad-CAM on the Potsdam dataset.

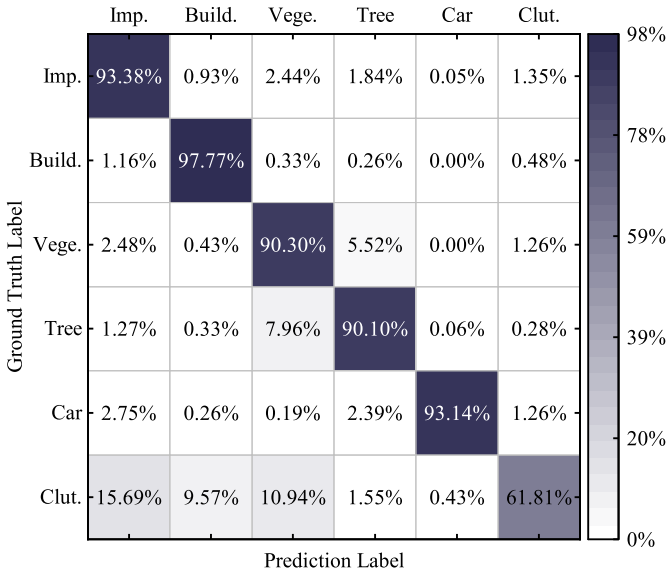


Fig. 9. Confusion matrix of semantic segmentation results on the Potsdam dataset.

precise and concentrated activations that are well aligned with object boundaries, while effectively suppressing irrelevant background responses. This leads to clearer delineation of object contours, which is particularly important for remote sensing segmentation.

11) Study on Confusion Matrix: We further analyze the segmentation performance using the confusion matrix, as shown in Fig. 9. Overall, most categories achieve high classification accuracy, as indicated by the strong responses along the diagonal. In particular, *building* and *impervious surface* are well recognized due to their relatively regular structures and large spatial extents. However, several challenging class pairs still exhibit noticeable confusion. In particular, significant confusion is observed between *vegetation* and *tree*, with 5.52% of vegetation pixels misclassified as tree and 7.96% in the opposite direction. This is mainly due to their similar spectral characteristics and overlapping spatial patterns. In addition, *car* shows confusion with surrounding categories such as impervious surfaces and trees, which can be attributed to its small object size and sensitivity to spatial resolution. The most challenging category is *clutter*, which shows a relatively low accuracy of 61.81%. A large portion of clutter pixels are misclassified as impervious surfaces, buildings, and vegetation. This is expected since clutter typically contains heterogeneous and ambiguous regions without clear semantic boundaries.

12) Performance-Speed Tradeoff Ablation: As shown in Table XII, we compare computational cost and segmentation performance in terms of trainable parameters, peak GPU memory, training time, inference latency, FLOPs, FPS, and mIoU under identical experimental settings. Compared with general segmentation methods, PEFT approaches significantly reduce the number of trainable backbone parameters while maintaining competitive segmentation accuracy. MsRE achieves 80.98% mIoU with only 4.64M trainable backbone parameters, showing strong parameter efficiency. Compared with LoRA and AdaptFormer, MsRE obtains the highest mIoU among the compared PEFT methods. However, MsRE also requires higher peak GPU memory, longer training time, and larger inference latency than LoRA and AdaptFormer. Therefore, MsRE is not the fastest PEFT method. The additional cost mainly comes from the multiscale cognitive operations and token interactions, which introduce extra feature-refinement computation. These modules improve scale-aware representation learning, but they inevitably increase memory usage and latency. Overall, the results indicate that MsRE provides a favorable accuracy-efficiency tradeoff rather than an absolute latency advantage: it achieves the best accuracy among the compared PEFT methods with a small trainable-parameter budget, at the cost of moderate additional memory and runtime overhead.

V. Discussion

A. Discussion on MsRE

The effectiveness of MsRE can be attributed to its ability to explicitly model multiscale feature interactions. Remote sensing imagery inherently contains objects with large variations in scale, making single-scale feature representations insufficient. By introducing multiscale visual cognitive operations, MsRE captures features at different cognitive fields, enabling richer representations. After that, the token interaction mechanism provides an effective way to align and refine features across different scales. The use of shared learnable tokens facilitates cross-scale information exchange, leading to more consistent and discriminative representations. This is also validated by our ablation studies, where replacing shared tokens with scale-specific tokens leads to degraded performance.

B. Compare With Existing PEFT Methods

Unlike conventional PEFT methods such as LoRA and Adapter, which mainly focus on parameter-efficient linear transformations, MsRE introduces a multiscale refinement mechanism tailored for remote sensing segmentation. In particular, MsRE enhances the backbone by incorporating multiscale

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TABLE XII
COMPUTATIONAL COST AND PERFORMANCE COMPARISON

Method	Backbone	Params†	Mem.(GB)↓	Train(s/iter)↓	Latency(s/iter)↓	Flops↓	FPS↑	mIoU%
General Segmentation Methods								
DeepLabv3+ [17]	Resnet-50	25.2 M	2.31	0.058	0.009	102 G	105.7	73.43
Segmenter [23]	ViT-Large	304.3 M	11.53	0.153	0.039	362 G	25.88	78.24
PEFT Methods								
LoRA [30]	DINOv3 (ViT-Large)	9.43 M	4.42	0.086	0.035	344 G	28.55	79.67
AdaptFormer [32]		3.12 M	4.08	0.078	0.036	347 G	28.92	79.34
MsRE (ours)		4.64 M	5.23	0.107	0.043	357 G	22.84	**80.98**

The experimental results were obtained using the Potsdam dataset on an NVIDIA RTX 4090D. Peak GPU memory, training time, inference latency, FLOPs, FPS, and mIoU were measured under identical hardware and experimental settings. Mark† denotes the number of trainable parameters in the backbone.

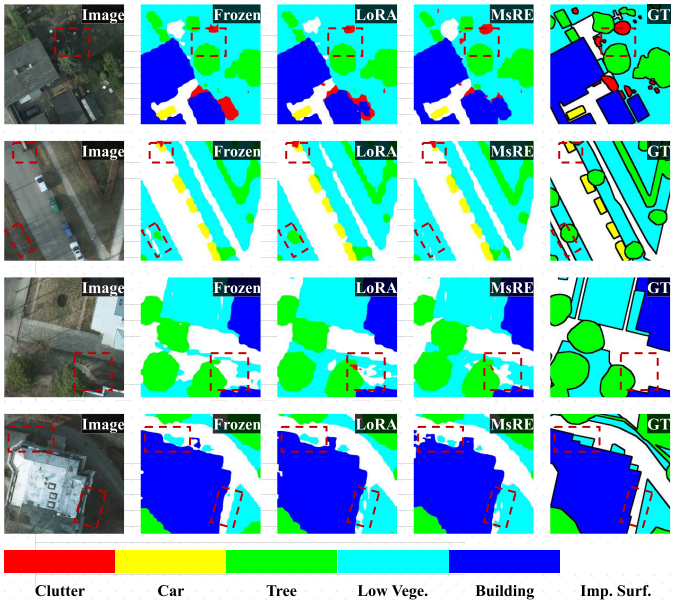


Fig. 10. Failure-case visualization on the Potsdam dataset. Red boxes highlight regions where MsRE still produces inaccurate predictions.

feature processing and spatially aware token interactions. This design makes MsRE more suitable for remote sensing tasks, where spatial structures and scale variations play a critical role. As a result, MsRE achieves a better balance between parameter efficiency and representation capability.

C. Failure Case

Although MsRE improves the overall segmentation quality, the failure cases in Fig. 10 shows that several challenging scenarios remain unresolved. First, small vehicles are still difficult to recognize reliably. They occupy only a few pixels and are easily confused with surrounding impervious surfaces or nearby trees, especially when object boundaries are weak or partially occluded. This observation is consistent with the classwise IoU comparison, where MsRE is slightly lower than FSegNet on the *car* category on Potsdam and remains clearly lower on Vaihingen. Second, clutter-like regions and ambiguous impervious surfaces remain challenging because they often contain heterogeneous visual patterns and unclear semantic boundaries. Even with multiscale feature refinement, these regions can be confused with buildings, vegetation, or impervious surfaces. Third, boundaries between *tree* and *low vegetation* are still occasionally confused due to their similar spectral characteristics and overlapping spatial structures. These visual failure cases are consistent with the confusion-matrix analysis, where noticeable confusion remains between vegetation and tree, and clutter shows relatively low recognition accuracy.

D. Limitations and Future Works

Despite its effectiveness, MsRE has some limitations. The introduction of multiscale cognitive operations and token interactions leads to increased computational overhead, resulting in lower inference speed compared to some weight update PEFT methods, such as LoRA. Nevertheless, MsRE requires far fewer trainable backbone parameters than full fine-tuning while achieving comparable segmentation accuracy. In addition, MsRE still has limitations in handling extremely small objects, ambiguous clutter regions, vegetation/tree confusion, and complex domain-shift scenarios. Future work may incorporate small-object enhancement, boundary-aware refinement, and domain adaptation to further improve robustness in these difficult cases and focus on developing more efficient cross-scale interaction strategies. Extending MsRE to other dense prediction tasks, such as object detection and change detection, is also a promising direction.

VI. CONCLUSION

Existing remote sensing segmentation models typically adopt relatively lightweight backbones, which limit their feature extraction capability. On the contrary, VFMs, pretrained on large-scale datasets, demonstrate significantly stronger representation capacity. However, the large number of parameters in VFMs makes full fine-tuning computationally expensive and inefficient when applied as segmentation backbones. Although existing PEFT methods alleviate this issue to some extent, they generally fail to effectively capture the inherent multiscale characteristics of remote sensing images. To address this limitation, this article proposes a novel method specifically designed for efficient remote sensing segmentation, termed MsRE tuning. MsRE enables efficient adaptation of VFMs to remote sensing segmentation tasks by introducing only a small number of trainable parameters. In the experiments, we first compare MsRE with existing segmentation methods and

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representative PEFT approaches, demonstrating its competitive effectiveness and parameter-efficient adaptation ability. We further conduct extensive ablation studies, including prediction and feature visualization analyses, and investigate the generalization capability of MsRE across different backbones and decoder heads. Finally, we analyze the computational efficiency of the proposed method. Overall, the experimental results verify that MsRE is an effective, robust, and efficient approach, providing a practical solution for leveraging VFMs in remote sensing segmentation.

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